



Unit 04: Project Planning, Scheduling and Monitoring

Topics to be covered:

- 4.1 Introduction
- 4.2 Need for Project Planning
- 4.3 Functions of Project Planning
- 4.4 Steps in Project Planning
- 4.5 Project Planning Structure
- 4.6 Tools of Project Planning
- 4.7 Project Scheduling
- 4.8 Problems in Scheduling Real-life Projects
- 4.9 Introduction to project monitoring
- 4.10 Setting Goals and Objectives
- 4.11 Introduction to Implementation
- 4.12 Project Evaluation and its importance
- 4.13 Challenges in Monitoring and Evaluation.
- 4.14 Case Studies

4.1 Introduction:

Project Planning, Scheduling, and Monitoring are important steps in managing a project successfully.

- **Project Planning** involves defining objectives, scope, resources and tasks to create a structured roadmap to complete project goals.
- **Project Scheduling** organizes these tasks into a timeline, setting dependencies and deadlines to ensure efficient workflow and timely completion.
- **Project Monitoring** continuously tracks progress, identifies deviations, avoids risks, and ensures the project stays aligned with the plan through regular assessments and adjustments.

Together, these processes help in efficient project execution, resource management, and achieving desired outcomes.

4.2 Need for Project Planning:

- ☐ Project planning is thinking before doing.
- ☐ Project Planning is all about designing effective policies and methods in order to fulfil project deliverables or project scope or project objectives.

Need for project planning are as follows:

- To complete project at given cost.
- To complete project at given time.
- To completely define all the project work to project team.
- To reduce risk.
- To reduce uncertainty.
- To reduce wastages.
- To improve efficiency of the project work.
- To obtain a better understanding of the project objectives.
- To provide a basis understanding on monitoring and controlling project work.

4.3 Functions of Project Planning:

Following are the functions of project planning:

- Finding out what needs to be done (scope).
- Breaking tasks into smaller parts and assigning them (task identification using WBS).
- Deciding who and what is needed to do each task (resource allocation).
- Estimating how long each task will take (time estimation).
- Identifying and managing potential problems (risk management).
- Making sure the work meets quality standards (quality planning).
- Deciding how to communicate with everyone involved (communication planning).
- Planning for anything that needs to be purchased or hired (procurement planning).
- Making sure everything fits together well (integration).
- Keeping records of everything (documentation).

4.4 Steps in Project Planning:

Following are the steps in Project Planning:

- a) **Set Goals:** Define what the project aims to complete.
- b) **Identify Stakeholders:** List the people involved and their roles.
- c) **Break down Tasks:** Divide the project into smaller steps (WBS).
- d) **Estimate Budget & Resources:** Plan costs, materials and team needs.
- e) **Create a Schedule:** Set deadlines and order of tasks.
- f) **Calculate Risks:** Identify problems that may arise and solutions.
- g) **Plan Communication:** Decide how updates and reports will be shared.

- h) **Ensure Quality:** Set standards and review processes.
- i) **Get Approval:** Confirm the plan with key decision-makers.
- j) **Monitor Progress:** Track tasks and make adjustments as needed.

4.5 Project Planning Structure:

The structure of project planning can be broken down into four main phases:

Getting Started:

- ✓ Figure out what you want to achieve (goals).
- ✓ Identify who will be involved (stakeholders).
- ✓ Check if it's possible to do (feasibility).

Planning:

- ✓ Decide what needs to be done (scope).
- ✓ Break work into smaller tasks (WBS).
- ✓ Decide when things will happen (schedule).
- ✓ Figure out who will do what (resources).
- ✓ Think about what could go wrong and how to deal with it (risk).
- ✓ Decide quality of project.
- ✓ Decide how to communicate.
- ✓ Plan how to purchase or hire (procurement).
- ✓ Make sure everything fits together (integration).

Execution:

- ✓ Do the work according to the plan.
- ✓ Keep everyone informed (communication).
- ✓ Keep an eye on progress and fix problems as they come up (monitoring and control).
- ✓ Manage any changes to the plan.

Finishing Up:

- ✓ Get formal agreement that the work is done (acceptance).
- ✓ Close the contracts with outside suppliers (procurement).
- ✓ Learn from positive and negative (lessons learned).
- ✓ Conduct a post project review.
- ✓ Hand over the deliverables to the customers or stakeholders.

4.6 Tools of Project Planning:

- ❑ Project planning involves various tools that help in organizing, scheduling and managing resources effectively.

Following tools are available for making project planning.

- Gantt Chart
- Bar Chart
- Flow Chart
- WBS – (Work Breakdown Structure)
- CPM – (Critical Path Methods) - Network Diagrams
- PERT – (Project Evaluation & Review Technique) - Network Diagrams
- Time Estimates
- Quality Management Tools
- Risk Management Tools
- Project Management Software
- Communication Plans
- Resource Allocation Tools
- SWOT Analysis – (Strength Weakness Opportunity & Threat)
- Mind Mapping
- Project Schedules
- Feasibility Studies
- Budgeting and Cost Estimation Tools
- Decision Matrices
- Procurement Plans
- Stakeholder Analysis and Engagement Tools

Q. The bus manufacturing company produces Hi-tech sleeper AC bus; show the different activities schedule in the Gantt chart. (Dec-2023)

The Bus manufacturing company have following activities:

- Job 1 - Market Research & Customer Requirements
- Job 2 - Design & Engineering
- Job 3 - Procurement of Materials & Components
- Job 4 - Manufacturing Body Structure
- Job 5 - Installation of Sleeper Features
- Job 6 - Installation of AC System
- Job 7 - Quality Control & Testing
- Job 8 - Final Assembly & Inspection
- Job 9 - Marketing & Sales Launch
- Job 10 - Post-Sales Support & Customer Feedback

Q. The manufacturing company produces desktop computers, show the different activities schedule in the network chart. (Dec-2023)

The desktop manufacturing company may have following activities:

- ✓ Manufacture / Purchase Components/ Tools Required:
 - Monitor
 - Mouse
 - Keyboard
 - Mother board
 - CPU
 - Fan
 - Random Access Memory (RAM)
 - Hard Drive /disk
- ✓ Assembling of all above components.
- ✓ Installing Motherboard, RAM, I/O devices, hard drive, power supply.
- ✓ Testing
- ✓ Packaging

4.7 Project Scheduling:

- ☐ Project Scheduling is a procedure of identifying tasks or activities so that it can be completed within fixed cost and time.
- ☐ Scheduling shows, when to start and how much to be done.
- ☐ Scheduling usually consists of the planned start date and finish date.
- ☐ Scheduling is a critical component for effective time management in a project.

Purpose of project scheduling is as follows:

- i. **Planning:** Helps in managing tasks and resources effectively.
- ii. **Time Management:** Ensures tasks are completed on time.
- iii. **Coordination:** Helps in coordination among team members.
- iv. **Identifying Dependencies:** Helps in understanding task dependencies.
- v. **Risk Management:** Helps in identifying potential risks.
- vi. **Communication:** Clear communication with stakeholders.
- vii. **Budgeting:** Proper budget planning and monitoring spending.
- viii. **Tracking Progress:** Monitoring the progress of the project.
- ix. **Achieving Goals:** Completing the project successfully.

4.8 Problems in Scheduling Real-life Projects:

Here are some common problems in scheduling real-life projects:

Not Enough Workers or Equipment:

- The right people are not available when needed.
- Machines or materials are not available when needed, causing delays.
- Meeting hall or workspaces are not free when required.

Unrealistic Deadlines:

- The project timeline is too short, making it hard to finish on time.
- People have to rush, leading to mistakes and low-quality work.
- There is no extra time planned for unexpected problems.

Tasks That Depend on Each Other:

- Some tasks must be done in order, like building walls only after laying the foundation.
- If one task is delayed, the whole project gets delayed.

Not Enough Money:

- The budget is too small, so work stops due to insufficient funds.
- Costs of project are higher than expected, forcing to change schedules.

Changing Project Requirements:

- The client or management keeps adding new tasks.
- Frequent changes make it hard to follow the original schedule.

Bad Weather or Natural Events:

- Rain, storms, or extreme heat causes delay of work like construction work.
- Natural disasters stop work or damage materials.

Technical Problems:

- Computers, software, or machines stop working, delaying progress.
- Engineers or IT teams need extra time to fix issues.

Government Rules and Approvals:

- Getting permits or legal approvals take longer than planned.
- New rules force the project to change its schedule.

Poor Communication:

- Teams don't share updates, leading to confusion.
- Misunderstandings cause delays and extra work.

Unexpected Problems:

- Workers get sick or quit suddenly.
- Supplies don't arrive on time due to global issues like pandemics or wars.

4.9 Introduction to project monitoring

- ☐ Project monitoring is the process of tracking a project progress to make sure that the project is completed on fixed time and cost also meets its objectives.
- ☐ It helps project manager to identify issues early and take corrective actions before they become major problems.

Why is Project Monitoring Important?

- Keeps the Project on Track:** Ensures tasks are completed on time.
- Controls Costs:** Helps prevent overspending by monitoring the budget.
- Improves Quality:** Ensures work meets required standards.
- Manages Risks:** Identifies and resolves potential problems early.
- Improves Communication:** Provides updates to stakeholders and the team.

Key Features of Project Monitoring

- Time Monitoring:** Tracking if tasks are being completed on schedule.
- Cost Monitoring:** Ensuring spending stays within the budget.
- Quality Monitoring:** Checking that work meets expectations.
- Risk Monitoring:** Identifying and managing potential risks.
- Resource Monitoring:** Making sure people and materials are being used efficiently.

How is Project Monitoring Done?

Project monitoring is done by regularly checking progress and making sure everything is going as planned.

- **Regular Updates:** Teams give reports on what has been done and any problems they face.

- **Meetings:** Managers and team members discuss progress and find solutions to issues.
- **Tracking Tools:** Software like charts, dashboards or checklists help keep track of tasks.
- **Key Performance Indicators:** Measuring success using specific goals.

4.10 Setting Goals and Objectives:

Setting Goals and Objectives in a Project

- ☐ Setting clear goals and objectives is important for project success.
- ☐ They help guide the team, measure progress and ensure the project delivers the expected results.

Difference between Goals and Objectives

Goals:

- The overall purpose of the project.
- They are broad and long-term.
- Example: "Improve school facilities for better student learning."

Objectives:

- Specific, measurable steps to achieve the goal.
- They are detailed and short-term.
- Example: "Renovate five classrooms with new desks and whiteboards within three months."

How to Set Good Goals and Objectives? (SMART Method)

S – Specific:

- Be clear about what needs to be done.
- **Example:** "Improve student's math test scores by providing extra class."

M – Measurable:

- Able to measure and by when it can be completed.
- **Example:** "Increase the average test scores from 50% to 85% by the end of the semester."

A – Achievable:

- Should be realistic with fixed time and resources.

- **Example:** "Take two hours of extra class per week for three months."

R – Realistic:

- Should match objective.
- **Example:** "Conduct the class in batches to improve performance and build confidence."

T – Time-bound:

- Must have fixed time or deadline to complete.
- **Example:** "Achieve the test score improvement by the end of the semester."

Why Setting Goals and Objectives is Important?

- ✓ Keeps the team focused.
- ✓ Helps track progress and success.
- ✓ Improves decision-making and resource management.
- ✓ Ensures the project meets its purpose.

4.11 Introduction to Implementation

- ☐ **Project implementation** is the phase where plans turn into action.
- ☐ It is a crucial stage where the actual work happens.
- ☐ Proper planning, monitoring and teamwork help ensure successful execution and project completion.
- ☐ It involves executing tasks, managing resources and ensuring that the project meets its goals within the planned time and budget.

Why is Implementation Important?

- To make sure that project plans are carried out effectively.
- It helps in achieving project goals and deliver results.
- Helps in Identifying and resolving problems during project execution.

Key Steps in Project Implementation

- **Start the Work:** Assign tasks and begin project activities.
- **Manage Resources:** Use people, materials and money wisely.
- **Monitor Progress:** Track tasks to ensure they stay on schedule.
- **Solve Problems:** Solve problems and adjust plans if needed.
- **Communicate Regularly:** Keep the team and stakeholders updated.

- **Ensure Quality:** Check that the work is done as per required standards.

Challenges in Project Implementation

- Delays due to unexpected problems.
- Budget overruns if costs are not controlled.
- Poor communication leading to misunderstandings.
- Resource shortages affecting progress.

4.12 Project Evaluation and its importance.

- ☐ Project Evaluation is a step-by-step process of collecting and recording information about project results.
- ☐ Project evaluation is the process of checking whether a project is successful and meets its goals.
- ☐ It measures how well the project is planned, executed and completed.
- ☐ Project evaluation includes short-term and long-term project outcomes.

Evaluation can be done:

- Before the project starts:** To check if the project is possible (Feasibility Study).
- During the project:** To monitor progress and make improvements (Ongoing Evaluation).
- After the project is completed:** To measure success and learn for future projects (Final Evaluation).

Why is Project Evaluation Important?

- It checks whether the goals are met or not.
- It helps in better planning and execution.
- It helps to track progress.
- It reduces risks.
- It improves efficiency.
- It helps in decision-making.
- It helps in proper use of resources.
- It helps in future project.
- It helps to find success factors.
- It helps team to learn, improve and innovate.
- It helps to measures long-term success

4.13 Challenges in Monitoring and Evaluation.

Monitoring and evaluation help track a project's progress and success. Following are the several challenges in monitoring and evaluation:

- i) **Unclear Goals:** If project goals are not clear, measuring success is hard.
- ii) **Poor Data:** Missing or incorrect data affects accuracy.
- iii) **Lack of Resources:** Not enough time, money or skilled staff.
- iv) **Unwilling to Share Feedback:** Some people may not want to give honest opinions and feedback.
- v) **Frequent Changes in scope:** If the project keeps changing its scope, tracking progress becomes difficult.
- vi) **Unfair Reports:** Some data may be changed to look better than reality.
- vii) **Hard to Measure Impact:** Some project benefits take a long time to show.
- viii) **Poor Communication:** Lack of teamwork and coordination causes problems.

How to solve challenges in Monitoring and evaluation?

- Set clear goals so success is easy to measure.
- Collect reliable data and check its accuracy.
- Ensure enough resources like time, money and skilled staff.
- Encourage honesty in reporting and feedback.
- Adapt to changes while keeping track of progress.

By fixing these issues, monitoring and evaluation become more effective and help projects success.

4.14 Case Studies.

Case Study 3: Assume that you are organizing a one-day intercollege student Debate competition at your institution.

Requirements:

- 2 participants and 1 staff member from about 50 colleges across state.
- Judges are to be identified.
- Budget plan for Food, Accommodation and Prizes has to be made.
- Stage and seating arrangement should be considered.

Task:

- Estimate (Plan) the budget for the entire activity

- Prepare the schedule chart for the entire activity (any planning tools can be used).
- Give a detailed monitoring report as per the scheduled chart.
- Present it in a seminar, with each group getting 5-10 minutes to present their idea.

Case Study 3 Solution:

Case Study 3: Organizing a One-Day Intercollege Student Debate Competition

1. Introduction

An intercollege student debate competition is being organized at our institution, with participation from 50 colleges across the state. Each college will send 2 participants and 1 staff member, making a total of 150 attendees. The event requires detailed planning for judging, budgeting (food, accommodation, prizes), and logistics (stage, seating, and scheduling).

2. Work Breakdown Structure (WBS)

Debate Competition Organization

- 1.1 Pre-Event Planning
 - Define objectives & budget
 - Obtain approvals & permissions
 - Identify judges & confirm availability
 - Risk assessment
- 1.2 Logistics & Arrangements
 - Venue selection & stage setup
 - Seating & sound system arrangement
 - Registration & ID allocation
- 1.3 Budget Planning
 - Food (Lunch & Refreshments)
 - Accommodation for outstation participants
 - Prizes & awards
- 1.4 Event Execution
 - Opening ceremony & speeches
 - Debate competition rounds
 - Breaks & refreshment timing
 - Closing ceremony & prize distribution
- 1.5 Post-Event Closure

- Feedback collection
- Financial reconciliation
- Report submission

Debate Competition Organization

1.1 Pre-Event Planning

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1.4 Event Execution

- Opening ceremony & speeches
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1.5 Post-Event Closure

- Feedback collection
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3. Budget Estimation

Expense Category	Estimated Cost per Person (INR)	Total Cost (for 150 People) (INR)
Food (Lunch & Snacks)	₹ 500	₹ 75,000
Accommodation (for outstation participants)	₹ 1,200	₹ 1,80,000
Prizes & Trophies	₹ 50,000	₹ 50,000
Stage Setup & Seating	₹ 70,000	₹ 70,000
Sound System & Lighting	₹ 30,000	₹ 30,000
Miscellaneous (Certificates, Decorations, etc.)	₹ 25,000	₹ 25,000
Total Estimated Cost		₹ 4,30,000

4. Schedule Chart (Event Timeline)

Expense Category	Estimated Cost per Person (INR)	Total Cost (for 150 People) (INR)
Food (Lunch & Snacks)	₹ 500	₹ 75,000
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Total Estimated Cost		₹ 4,30,000

5. Monitoring Report

Monitoring Aspect	Responsibility	Status
Registration Process	Event Committee	☑ Completed
Judge Arrangements	Debate Coordinator	☑ Completed
Stage & Sound Setup	Logistics Team	☑ Ready
Food & Refreshments	Catering Team	☑ Arranged
Seating Arrangements	Logistics Team	☑ Set
Debate Competition Flow	Debate Moderators	In Progress
Prize Distribution	Awards Committee	☑ Pending
Feedback Collection	Event Volunteers	☑ Pending

A monitoring team will be assigned to oversee event execution and ensure smooth operation.

6. Seminar Presentation Plan**Presentation Structure (5-10 Minutes)**

1. Introduction (1 min) – Overview of the debate competition.
2. Event Plan (2 min) – WBS & budget breakdown.
3. Schedule & Logistics (3 min) – Detailed event timeline & execution.
4. Monitoring & Risk Management (2 min) – How we ensured smooth event flow.
5. Conclusion (2 min) – Key takeaways & open discussion.

Question bank (05 to 10 Marks):

1. Explain Project Planning, Scheduling and Monitoring.
2. What is the Need for Project planning?
3. What are the functions of Project Planning?
4. List the steps in Project Planning.
5. Explain Project Planning Structure.
6. List the tools used in Project planning (any ten).
7. What do you mean by Project Scheduling?
8. What is the purpose of Project Scheduling?
9. List the Problems in Scheduling Real-life Projects.
10. Explain Project Monitoring.
11. Why Project Monitoring is important?
12. Explain SMART in Project Goals and objectives.
13. Explain Project Implementation.
14. Why Project Implementation is important and what are the challenges in Project Implementation.
15. Explain Project Evaluation.
16. Why Project Evaluation is important?
17. What are the challenges in Project Monitoring and Evaluation?



